

Doc, a declarative PDF API

Doc describes a document as a list of typed blocks. You compose content with plain objects, set styles once, and call build. Inline runs mix **bold**, italic, a link, inline code like `doc.page()`, and math like $e^{i\pi} + 1 = 0$.

Getting started

Create a document, add a page, push blocks into content, then build to a file.

```
import { Doc } from "@nemu-ai/pdf";

const doc = new Doc({ page_size: "A4", margin: 54 });
const page = doc.page();
page.content(
  { type: "heading", text: "Hello" },
  { type: "paragraph", text: "First document." },
);
await doc.build("out.pdf");
```

Styling

set_style assigns defaults per block role. Any block can override with its own style. Colors accept a string or a Color value.

```
doc.set_style({
  heading: { font_family: "inter" },
  paragraph: { font_size: 12, line_height: 1.6 },
  formula: { font: "termes", font_size: 13 },
});
```

Headings and lists

Headings take a level from 1 to 6. Lists are ordered or unordered, and items accept inline content.

```
page.content(
  { type: "heading", level: 2, text: "Topics" },
  { type: "list", ordered: true, items: ["First", "Second"] },
);
```

Text, code, and math

Code blocks

Code blocks render on a tinted background in a monospace font. Inline code sits in the text flow.

```
page.content({
  type: "code",
  language: "ts",
  text: "const x = 1;",
});
```

Formulas

Formulas render through MathJax. The math typeface is selectable with the font option, and you control the size.

```
doc.set_style({ formula: { font: "termes", font_size: 13 } });
page.content({ type: "formula", text: "x = \\frac{-b \\pm \\sqrt{b^2 - 4ac}}{2a}" });
```

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

Components

Note

A note draws a tinted callout with an optional title. Variants are info, warn, success, and muted.

```
page.content({  
  type: "note",  
  variant: "info",  
  title: "Tip",  
  text: "Notes group a short message.",  
});
```

Tip

Notes group a short message in a tinted box.

Careful

Use warn for cautions.

Table

Tables take a header row and data rows. Cells are plain strings or objects with an alignment.

```
page.content({
  type: "table",
  headers: ["Name", "Role", { text: "Score", align: "right" }],
  rows: [["Alice", "Lead", { text: "98", align: "right" }]],
});
```

Region	Plan	Users	Growth
North	Pro	1,284	14%
South	Team	892	9%
West	Pro	2,011	23%

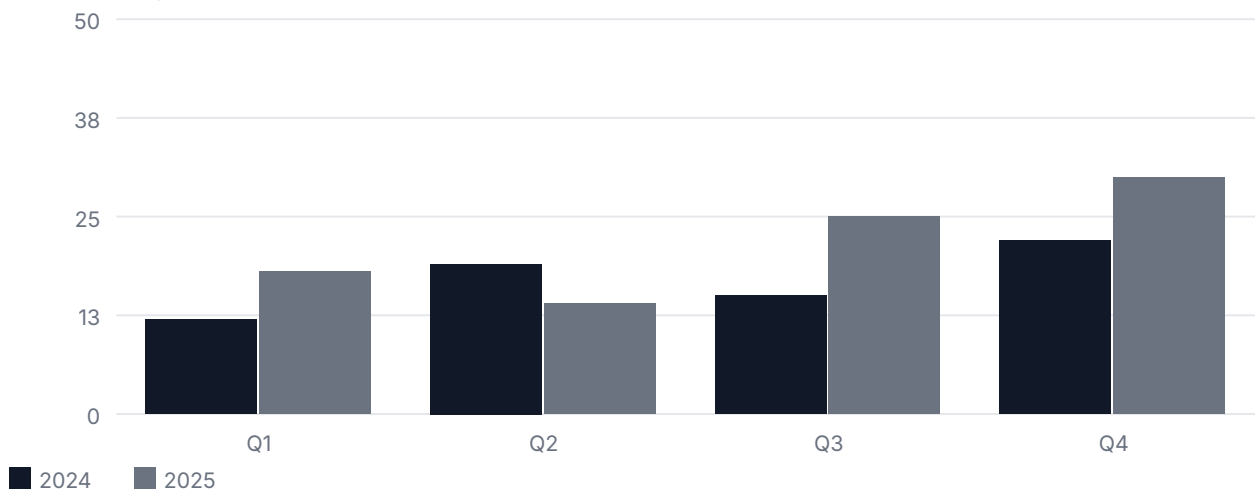
Charts

Charts render as native PDF vectors. Supported kinds are bar, line, area, pie, and donut.

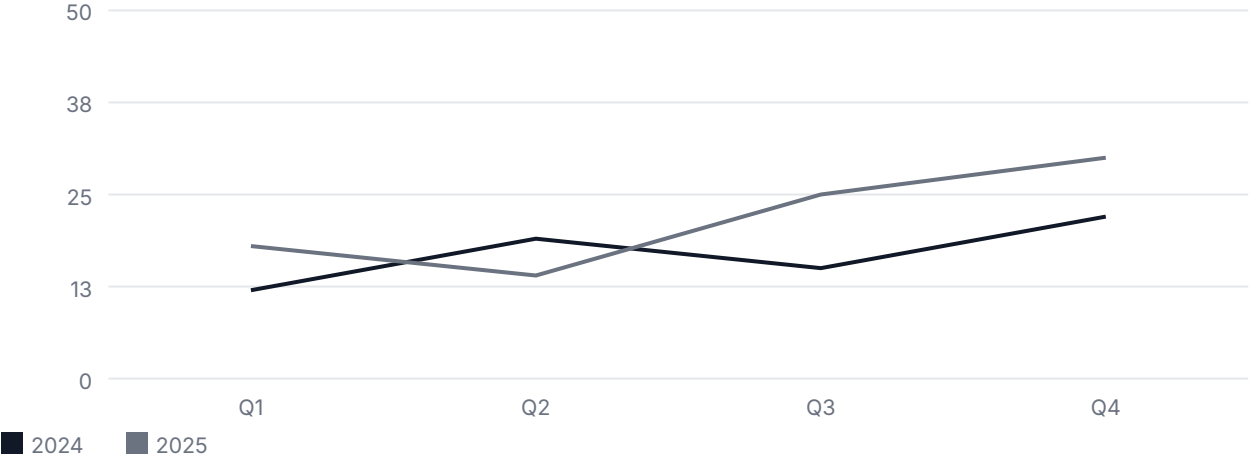
Series and slices accept Color values.

```
page.content({
  type: "chart",
  chart: "bar",
  title: "Revenue",
  data: { labels: ["Q1", "Q2"], series: [{ name: "2025", values: [18,
14] }] },
});
```

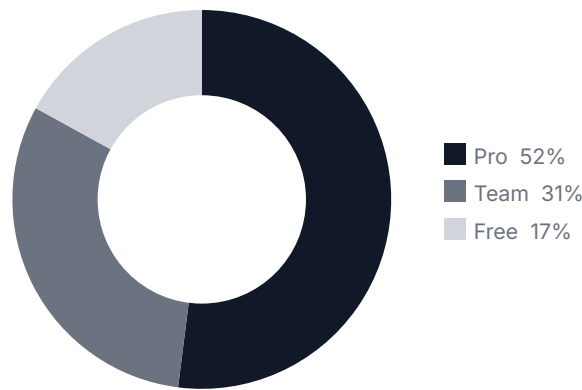
Revenue by quarter



Trend



Plan mix



Fonts and colors

Six fonts ship with the package and register on load. Reference any by name. You can also load your own font file.

```
doc.set_style({ paragraph: { font_family: "roboto" } });
doc.load_font("brand", "./fonts/brand.ttf");
```

- inter
- geist
- geist-mono
- nunito-sans
- roboto
- source-serif-4

Color

Color builds from one input and converts to any format. Pass a Color anywhere a color is accepted.

```
import { Color } from "@nemu-ai/pdf";

const c = Color({ hex: "#3366cc" });
c.to_rgb();      // rgb(51, 102, 204)
c.to_hsl();      // hsl(220, 60%, 50%)
c.darken(0.2);   // a darker Color
doc.set_style({ heading: { color: c } });
```